

# Assertion languages

with modalities and separating connectives

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# Agenda

- A short note about my team and hosting institution
- Motivation from the world of separation logic
- An introduction to  $\text{QCTL}_{\text{EX}}$
- Some toy formulae
- A new technique of local nominals
- Proof ideas
- Other results
- Our current research and open problems
- Summary

# Team

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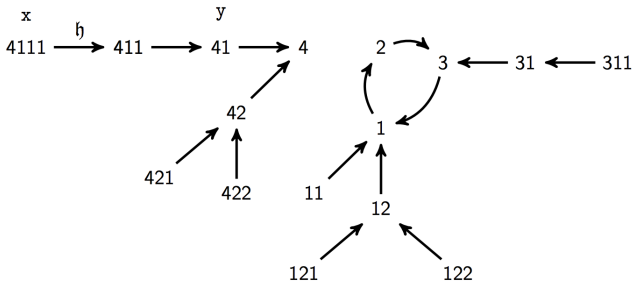
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# Motivation

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- **Modal Separation Logic (MSL)** - describes **properties of heaps**



- Navigation modalities  $\Diamond, \Diamond^{-1}, <\neq>$
- Separation logic features - star  $*$ , magic wand  $-*$

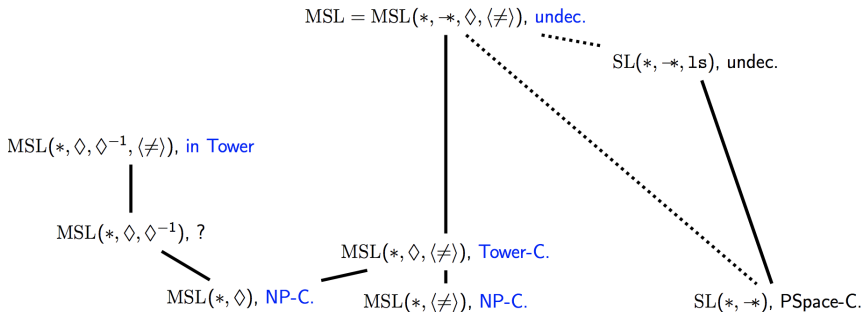
# Satisfiability problem

Satisfiability problem for a logic  $\mathcal{L} \subseteq \text{MSL}$ :

**input:** a formula  $\phi$  in  $\mathcal{L}$ .

**output:** 1 iff  $\phi$  is satisfiable.

- in our case  $\mathcal{L} = \text{MSL}(\Diamond^{-1}, *)$
- What is the exact complexity?
- Current status of some fragments - next page

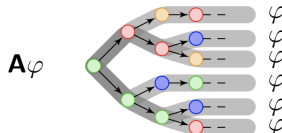
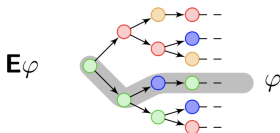


- PSpace-C. of  $SL(*, -*)$  [Calcagno & Yang & O'Hearn, FSTTCS'01]
- Undec. of  $SL(*, -, 1s)$  [Demri & Lozes & Mansutti, FOSSACS'18]
- Complexity class Tower [Schmitz, ToCT 2016]

# An introduction to QCTL

# Computation-Tree Logic (CTL)

- **atomic propositions:** , , ...
- **boolean combinators:**  $\neg\varphi$ ,  $\varphi \vee \psi$ ,  $\varphi \wedge \psi$ , ...
- **path quantifiers:**



- **temporal modalities:**

$X\varphi$    $\rightarrow$    $\rightarrow$    $\rightarrow$    $\rightarrow$    $\rightarrow$  ...

"next  $\varphi$ "

$\varphi U \psi$    $\rightarrow$    $\rightarrow$    $\rightarrow$    $\rightarrow$    $\rightarrow$  ...

" $\varphi$  until  $\psi$ "

$\text{true } U \varphi \equiv F\varphi$    $\rightarrow$    $\rightarrow$    $\rightarrow$    $\rightarrow$    $\rightarrow$  ...

"eventually  $\varphi$ "

$\neg F \neg\varphi \equiv G\varphi$    $\rightarrow$    $\rightarrow$    $\rightarrow$    $\rightarrow$    $\rightarrow$  ...

"always  $\varphi$ "

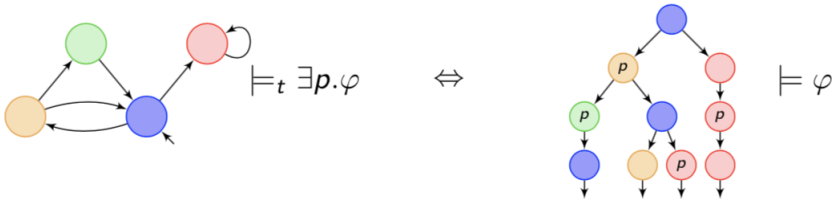
# Quantified CTL

[Kup95,Fre01]

QCTL extends CTL with **propositional quantifiers**

$\exists p. \varphi$  means that **there exists a labelling** of the model with  $p$  under which  $\varphi$  holds.

- tree semantics:



# Complexity results

Theorem (LMCS 2014, Laroussinie and Markey)

QCTL is Tower-hard already for binary trees

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QCTL is in Tower

- We consider  $\text{QCTL}_{\text{EX}}$  i.e. QCTL with X operator only.
- Such logic is expressive enough to capture  $\text{MSL}(\Diamond^{-1}, *)$
- $\text{QCTL}_{\text{EX}}$  is an important fragment of QCTL
- Probably more interesting that the whole separation logic stuff
- We hoped for elementary complexity.

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Proof extremely technical, we will not present it now.

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## Theorem (Bounded degree)

$\text{QCTL}_{\text{EX}}$  is in  $\text{STA}(*, 2^{\mathcal{O}(n)}, n)$  over trees with bounded degree and NExpTime-hard (we conjecture  $\text{STA}(*, 2^{\mathcal{O}(n)}, n)$ -completeness).

Hardness more interesting.

# Examples

## Useful properties

### Lemma

Every satisfiable  $\text{QCTL}_{\text{EX}}$  formula  $\varphi$  has a tree model of a depth bounded by a modal depth of  $\varphi$ .

### Lemma

Any  $\text{QCTL}_{\text{EX}}$  formula  $\varphi$  can be transformed into equisatisfiable formula in Prenex Normal Form, which length is bounded linearly in the size of  $\varphi$ .

Bottom-up rewriting system:

- $\mathcal{Q}x.\varphi \equiv \overline{\mathcal{Q}}x.\neg\varphi,$
- $\mathcal{Q}_1x_1.\varphi_1 \vee \mathcal{Q}_2x_2.\varphi_2 \equiv \mathcal{Q}_1x_1.\mathcal{Q}_2x_2.\varphi_1 \vee \varphi_2,$  and
- $\text{EX}(\mathcal{Q}x.\varphi) \equiv \mathcal{Q}x.\text{EX}(\varphi),$

# Counting abilities

## ■ Uniqueness

$$\text{EX}_{=1}(\varphi) \stackrel{\text{def}}{=} \forall p \text{EX}(\varphi) \wedge \neg (\text{EX}(p \wedge \varphi) \wedge \text{EX}(\neg p \wedge \varphi))$$

## ■ Graded modalities

$$\text{EX}_{\geq n}(\varphi) \stackrel{\text{def}}{=}$$

$$\exists p_0 \exists p_1 \dots \exists p_{n-1} \bigwedge_{i=0}^{n-1} \text{EX}(p_i \wedge \varphi) \wedge \bigwedge_{0 \leq i < j \leq n-1} \neg \text{EX}(p_i \wedge p_j \wedge \varphi)$$

# Hybrid logic

Hybrid logic extends modal logic with three following constructs:

- nominals (aka. constants)  $x, y, z, \dots$
- binders  $\downarrow$  (assigning name to a given state)
- at operator  $@$  (access to the state with a given name)

It is **not possible** to define them in  $\text{QCTL}_{\text{EX}}$  since **depth** of models is **bounded**. We define something similar called **local nominals**.

# Local nominals

- An ability to **quantify over nodes** in the **limited scope**
- We will use binders  $\downarrow_x^k$  to **set up a uniquely identified node** labeled with  $x$  exactly  $k$  steps below. To **access such a node** we will use a macro  $@_x^k$

$$\begin{aligned}\downarrow_x^1 & \stackrel{\text{def}}{=} EX_{=1}(x), \quad @_x^1(\varphi) \stackrel{\text{def}}{=} EX(x \wedge \varphi) \\ \downarrow_x^{k+1} & \stackrel{\text{def}}{=} \exists \text{pathto}_x \downarrow_{\text{pathto}_x}^1 \wedge @_x^1(\varphi) \downarrow_x^k, \\ @_x^{k+1}(\varphi) & \stackrel{\text{def}}{=} @_x^1(\varphi) \downarrow_x^k\end{aligned}$$

# Local nominals - example

To bind **two different** local nominals **at the same level** we use:

$$\downarrow_{x,y}^k \stackrel{\text{def}}{=} \downarrow_x^k \wedge \downarrow_y^k \wedge @_x^k(\neg y) \wedge @_y^k(\neg x)$$

Example: **every successor** is **labeled** with a **different subset** of AP

$$\varphi_{\text{uniqueAP}} \stackrel{\text{def}}{=} \underbrace{\forall x \forall y \downarrow_{x,y}^1 \rightarrow}_{\text{name two different successors}} \neg \left( \underbrace{\bigwedge_{a \in \text{AP}} @_x^1(a) \leftrightarrow @_y^1(a)}_{\text{agree on AP}} \right)$$

# Proof ideas

# Number encodings

- We use **unary predicates**  $a_i$  to denote **binary encodings** of natural numbers.
- $A_n$  - predicates for  $n$  bits
- $A_4=5$  will be expanded as  $\neg a_3 \wedge a_2 \wedge \neg a_1 \wedge a_0$ .
- We would like to employ arithmetical operations like  $=$ ,  $+1$  and  $>$  between the numbers represented by two local nominals

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$$\varphi_{=(x, y, \text{lvl}, A_n)} \stackrel{\text{def}}{=} \bigwedge_{i=0}^{n-1} @_x^{\text{lvl}}(a_i) \leftrightarrow @_y^{\text{lvl}}(a_i).$$

# Plus one relation

$$\bigwedge_{i=0}^{n-1} \left( \underbrace{@_x^{lvl} \left( \neg a_i \wedge \bigwedge_{j=0}^{i-1} a_j \right)}_{\text{look for the first zero bit}} \rightarrow \left( \underbrace{@_y^{lvl} \left( \bigwedge_{j=0}^{i-1} \neg a_j \wedge a_i \right)}_{\text{reset previous bits, set } a_i} \wedge \underbrace{\bigwedge_{j=i+1}^{n-1} @_x^{lvl}(a_j) \leftrightarrow @_y^{lvl}(a_j)}_{\text{rewrite other bits}} \right) \right)$$

# Greater than relation

$$\varphi_{>}(x, y, \text{lvl}, A_n) \stackrel{\text{def}}{=} \bigvee_{i=0}^{n-1} \left( @_x^{\text{lvl}}(a_i) \wedge @_y^{\text{lvl}}(\neg a_i) \wedge \underbrace{\bigwedge_{j=i+1}^{n-1} @_x^{\text{lvl}}(a_j) \leftrightarrow @_y^{\text{lvl}}(a_j)}_{\text{rewrite other bits}} \right)$$

# Bounded satisfiability

We are going to prove the following theorem:

## Theorem

The satisfiability problem for  $\text{QCTL}_{\text{EX}}$  over binary trees is NExpTime-hard.

An ExpSpace upper bound comes trivially (check all possible models). It contrasts with Tower-hardness for the whole logic.

## NExpTime tiling

A tiling system is a tuple  $S = (T, H, V)$ , where  $T$  is a finite set of tiles,  $H \subseteq T \times T$  and  $V \subseteq T \times T$  are, respectively, the horizontal and vertical adjacency relations.

A tiling problem instance is a tuple

$(S, n, \text{top}_0, \text{top}_1, \dots, \text{top}_{n-1}, \text{bot})$ , where  $n \in \mathbb{N}$  is a natural number,  $\text{top}_i \in T$  are initial tiles and  $\text{bot} \in T$  is a final tile. A square tiling of a grid  $G = 2^n \times 2^n$  is a mapping  $\tau : G \rightarrow T$ . A proper square tiling is a tiling, where mapping  $\tau$  respect the horizontal and vertical rules.

In square tiling problem we ask whether there exists a proper square tiling of a grid  $2^n \times 2^n$ , such that first  $n$  tiles in the top row are  $\text{top}_0, \text{top}_1, \dots, \text{top}_{n-1}$  (in the same order) and the tile  $\text{bot}$  appear somewhere at the bottom row.

## Defining grid

- To encode  $2^n \times 2^n$  grid we need  $2^n \times 2^n$  cells
- Each cell - leaf
- Coordinates  $H_n$  and  $V_n$  predicates

$$\varphi_{\text{grid}} \stackrel{\text{def}}{:=} \exists V_n \exists H_n \varphi_{\text{firstCell}} \wedge \varphi_{\text{vSuccessors}} \wedge \varphi_{\text{hSuccessors}} \wedge \varphi_{\text{uniqueness}}$$

- To define the first cell we simply put a nominal in the distance  $2n$  with value 0 on both coordinates.

$$\varphi_{\text{firstCell}} \stackrel{\text{def}}{:=} \exists x_{\text{fst}} \downarrow_{x_{\text{fst}}}^{2n} \wedge @_{x_{\text{fst}}}^{2n} (H_n=0 \wedge V_n=0)$$

$$\varphi_{\text{uniqueness}} \stackrel{\text{def}}{:=} \forall x \forall y \downarrow_{x,y}^{2n} \rightarrow \neg (\varphi_=(x, y, 2n, H_n) \wedge \varphi_=(x, y, 2n, V_n))$$

# Vertical and horizontal successors

$$\varphi_{\text{succH}}(x, y) \stackrel{\text{def}}{=} \varphi_{+1}(x, y, 2n, H_n) \wedge \varphi_{=}(x, y, 2n, V_n)$$

$$\varphi_{\text{succV}}(x, y) \stackrel{\text{def}}{=} \varphi_{+1}(x, y, 2n, V_n) \wedge \varphi_{=}(x, y, 2n, H_n)$$

$$\varphi_{\text{hSuccessors}} \stackrel{\text{def}}{=} \forall x \downarrow_x^{2n} \wedge @_x^{2n}(H_n \neq 2^n - 1) \rightarrow \exists y_h \downarrow_{y_h}^{2n} \wedge \varphi_{\text{succH}}(x, y_h)$$

$$\varphi_{\text{vSuccessors}} \stackrel{\text{def}}{=} \forall x \downarrow_x^{2n} \wedge @_x^{2n}(V_n \neq 2^n - 1) \rightarrow \exists y_v \downarrow_{y_v}^{2n} \wedge \varphi_{\text{succV}}(x, y_v)$$

# Encoding tiles for a given instance I

$$\varphi_{\text{tiles}} \stackrel{\text{def}}{=} \exists T_{|T|} \forall x \downarrow_x^{2n} \rightarrow @_x^{2n} \left( \bigvee_{t \in T} t \wedge \bigwedge_{t, t' \in T} (\neg t \vee \neg t') \right)$$

$$\varphi_{\text{init}} \stackrel{\text{def}}{=} \bigwedge_{i=0}^{n-1} \exists x_i \downarrow_{x_i}^{2n} \wedge @_{x_i}^{2n} (H_n=0 \wedge V_n=i \wedge t_i)$$

$$\varphi_{\text{bot}} \stackrel{\text{def}}{=} \exists x_{\text{fin}} \downarrow_{x_{\text{fin}}}^{2n} \wedge @_{x_{\text{fin}}}^{2n} (H_n=2^n-1 \wedge \text{bot})$$

$$\varphi_{hRules} \stackrel{\text{def}}{=} \forall x \downarrow_x^{2n} \wedge @_x^{2n} (H_n \neq 2^n-1) \rightarrow \exists y_h \downarrow_{y_h}^{2n} \varphi_{\text{succ}H}(x, y_h) \wedge \bigvee_{(t, t') \in H} (@_x^{2n}(t) \wedge @_{y_h}^{2n}(t'))$$

$$\varphi_{vRules} \stackrel{\text{def}}{=} \forall x \downarrow_x^{2n} \wedge @_x^{2n} (V_n \neq 2^n-1) \rightarrow \exists y_v \downarrow_{y_v}^{2n} \wedge \varphi_{\text{succ}V}(x, y_v) \wedge \bigvee_{(t, t') \in V} (@_x^{2n}(t) \wedge @_{y_v}^{2n}(t'))$$

# Conclusions

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$\text{QCTL}_{\text{EX}}$  is Tower-hard (thus complete).

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Open problems:

- Expressive power?
- Exact complexity for each alternation level
- Complexity of MSL?